Midterm Research Report

Fixed Graphs, Gluing Structures, and Topological Information

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Abstract

This report summarizes my study of Cayley graph spectra, surface embeddings, and combinatorial constructions of topological spaces. Three main references provide complementary tools: representation theory explains spectral information, rotation systems describe surface embeddings, and spatial graph constructions address geometric realizability. More recent reading motivates local link validation and boundary-state representations for tetrahedral gluings. The current research direction is to determine what topological information can vary when the underlying face-pairing graph remains fixed.

1 Research Development

The project began with finite groups and Cayley graphs, then developed toward face systems, surface topology, and different notions of equivalence.

The main conceptual progression is

$$\text{finite groups} \longrightarrow \text{Cayley graphs} \longrightarrow \text{face systems} \longrightarrow \text{topological spaces}.$$

The central observation is that these levels contain different information. A group and generating set determine a graph, but the graph does not determine which cycles are filled as faces. Likewise, an algebraic invariant need not determine the entire space.

My current question is therefore:

> When a combinatorial skeleton is fixed, how much topological variation can different gluing structures produce, and which invariants detect that variation?

2 Three Main References and Their Applications

2.1 Liu–Zhou: Spectra from Group Representations

Reference. Liu and Zhou, *Eigenvalues of Cayley graphs* [1].

Main idea

Let G be a finite group and let $S = S^{-1}$ be a generating set excluding the identity. Using right-multiplication steps, the adjacency operator is

$$(Af)(x) = \sum_{s \in S} f(xs).$$

Theorem 2.1 (Representation-theoretic decomposition)**.** *For each irreducible unitary representation ρ of G, define*

$$M_\rho = \sum_{s \in S} \rho(s).$$

The adjacency spectrum is obtained from the spectra of the matrices M_ρ, with each block spectrum repeated $\dim \rho$ *times.*

For abelian groups, every irreducible representation is a character, giving

$$\lambda_\chi = \sum_{s \in S} \chi(s).$$

For a k-regular Cayley graph,

$$L_0 = kI - A.$$

Application to my project

- Compute and check Cayley graph spectra using representation blocks rather than only a full adjacency matrix.
- Interpret spectral drawings through group symmetry and invariant eigenspaces.
- Use the ordinary graph spectrum as a baseline when comparing different face systems on the same graph.
- Compare ordinary graph operators with face-sensitive operators to identify which added structural information becomes detectable.

Proposition 2.2 (Ordinary spectral blindness to faces)**.** *Fix a graph and its edge weights. Changing its attached face system does not change its adjacency matrix or ordinary vertex Laplacian.*

Proof. Neither operator uses the attached faces in its definition. □

Limitation. Spectral coordinates do not automatically give an injective or crossing-free geometric embedding. When an eigenvalue is repeated, arbitrary truncation within its eigenspace can also destroy symmetry.

2.2 Brinkmann: Rotation Systems and Genus Search

Reference. Brinkmann, *A Practical Algorithm for the Computation of the Genus* [2].

Main idea

A rotation system specifies the cyclic order of incident darts at each vertex. Let α reverse darts and let σ encode these cyclic orders. With right-to-left composition, the cycles of

$$\phi = \sigma\alpha$$

describe the facial walks.

Theorem 2.3 (Rotation systems and Euler genus)**.** *For a finite connected graph with at least one edge, a rotation system determines an orientable cellular embedding satisfying*

$$|V| - |E| + F_\sigma = 2 - 2g_\sigma,$$

where F_σ is the number of facial walks. Consequently,

$$\gamma(\Gamma) = 1 + \frac{|E| - |V| - \max_\sigma F_\sigma}{2}.$$

Thus minimizing genus is equivalent to maximizing the number of faces.

Proposition 2.4 (Genus change under edge insertion)**.** *For an edge inserted between specified corners of existing vertices in a connected cellular embedding,*

$$\Delta g = \begin{cases} 0, & \text{if the corners belong to the same face,} \\ 1, & \text{if the corners belong to distinct faces.} \end{cases}$$

Proof. The insertion increases $|E|$ by one. It either splits one face into two or merges two faces into one. Euler's formula gives the result. □

Application to my project

- Replace visually selected cycles with explicit rotation data and reproducible face tracing.
- Enumerate different surface structures while keeping the Cayley graph fixed.
- Use Euler characteristic and valid pruning rules to organize finite searches.
- Separate uniform rotations from general local rotations, and distinguish a chosen embedding's genus from the graph's minimum genus.
- Treat equivalence checking as a separate step: equal genus does not imply equivalent cellular embeddings.

Limitation. An embedding provides an upper bound on minimum genus; minimality requires a matching lower bound or complete exclusion of smaller genera. Facial walks may repeat vertices or edges, so simple-face restrictions must be stated explicitly.

2.3 Barthel–Buccoliero: Geometric Realization

Reference. Barthel and Buccoliero, *Constructing embedded surfaces for cellular embeddings of leveled spatial graphs* [3].

Main idea

This paper begins with a graph already embedded in three-dimensional space. It asks whether the graph can lie on an embedded closed orientable surface whose complementary regions are open disks.

Theorem 2.5 (Four-level existence result)**.** *If a spatial graph admits a leveled embedding with at most four levels, then it admits a cellular embedding in a closed orientable surface embedded in* $\mathbb{R}^3$*.*

This restates Proposition 1 of the paper.

Application to my project

- Distinguish an abstract surface construction from its realization in three-dimensional space.
- Investigate whether a selected spatial realization of a Cayley graph admits a compatible cellular surface.
- Use cylinder and punctured-surface attachments as geometric models for changing the surrounding surface.
- Check cellularity using the current face boundaries, rather than relying only on the original drawing or spine.

Limitation. The construction concerns a specified spatial graph. It does not enumerate all abstract face systems or automatically produce a minimum-genus surface. Geometric realization remains an optional extension of the current finite gluing study.

3 Shared Mathematical Structure

The three references address different parts of the same construction process:

$$(G, S) \xrightarrow{\text{representation theory}} \text{graph operators},$$
$$\Gamma \xrightarrow{\text{rotation system}} \text{surface cellulation},$$
$$\Gamma \subset \mathbb{R}^3 \xrightarrow{\text{geometric construction}} \text{a compatible embedded surface}.$$

Their roles should not be confused:

- Spectral information describes an operator.
- Rotation data describes a combinatorial embedding.
- Spatial construction addresses geometric realizability.

3.1 Homology and Hodge Operators

For a finite real chain complex,

$$C_2 \xrightarrow{B_2} C_1 \xrightarrow{B_1} C_0, \qquad B_1 B_2 = 0,$$

define, using orthonormal bases,

$$L_1 = B_1^\top B_1 + B_2 B_2^\top.$$

Theorem 3.1 (Finite-dimensional Hodge correspondence)**.** *There is an isomorphism*

$$\ker L_1 \cong \ker B_1 / \operatorname{im} B_2.$$

Therefore,

$$\dim \ker L_1 = \dim H_1(C_\bullet; \mathbb{R}).$$

Proof. For every $x \in C_1$,

$$x^\top L_1 x = \|B_1 x\|^2 + \|B_2^\top x\|^2.$$

Hence $\ker L_1 = \ker B_1 \cap \ker B_2^\top$. Since $\operatorname{im} B_2 \subseteq \ker B_1$, orthogonal decomposition identifies this intersection with the homology quotient. □

For a closed connected orientable surface of genus g, this gives

$$\dim \ker L_1 = 2g.$$

This provides a connection between face data and spectral information. However, real homology does not record integral torsion, and Hodge spectra are not complete topological classifiers.

4 Local Validation and Boundary States

Recent reading on tetrahedral gluings [4] adds a useful methodological principle: a global construction should be checked through its local identifications and vertex links.

4.1 Local topology

A tetrahedral face pairing can create singularities, even when all intended faces have been identified. For a genuine triangulated 3-manifold, vertex links must have the appropriate local type:

$$\mathrm{Lk}(v) \cong \begin{cases} S^2, & \text{at an interior vertex,} \\ D^2, & \text{at a boundary vertex.} \end{cases}$$

This motivates checking links rather than inferring manifold structure from a drawing or face count.

4.2 Boundary-state representation

A partial construction can be summarized by

$$C(T) = \big(C_e(T), C_v(T)\big),$$

where C_e records boundary-edge identifications and orientations, and C_v records equivalence classes of boundary vertices.

The main methodological distinction is:

$$\text{enough information for future legality} \quad \not\Rightarrow \quad \text{enough information for topological classification.}$$

4.3 Application to my project

- Record local identifications explicitly and reject constructions with invalid links.
- Use boundary configurations to organize partial gluing histories.
- Preserve distinct histories initially, even when they share the same state.
- Compare their completed homology and topological invariants to determine what the compressed state has forgotten.

Theorem 4.1 (Admissibility and treewidth)**.** *For a connected 4-regular multigraph with n nodes and treewidth at most k, the existence of a closed 3-manifold triangulation with that face-pairing graph can be decided in $O(nf(k))$ time, given a suitable tree decomposition with $O(n)$ bags.*

This is an existence result from [4]. It does not directly provide an equally efficient enumeration or homeomorphism-classification algorithm.

5 Current Research Question

For a fixed face-pairing graph Γ, let $M_{\Gamma,\varphi}$ denote the space determined by gluing maps φ.

The main question is whether two gluing assignments can produce closed, connected, orientable 3-manifolds satisfying

$$H_j(M_{\Gamma,\varphi};\mathbb{Z}) \cong H_j(M_{\Gamma,\psi};\mathbb{Z}) \quad \text{for every } j,$$

while

$$M_{\Gamma,\varphi} \ncong_{\text{homeo}} M_{\Gamma,\psi}.$$

Related questions are:

- Can such a difference arise by changing only one face-gluing map?
- Can equal intermediate boundary states lead to different final homology?
- What additional information would a state need to retain a chosen topological invariant?

These are working research questions. Their novelty and answers have not been established.

6 Current Progress

- Prepared reading summaries connecting representation theory, rotation systems, spatial embeddings, and boundary-state methods.
- Rechecked an existing catalogue of 25 surface examples; all passed the current validator, including numerical Hodge checks.
- Refined the main direction toward fixed face-pairing graphs and topological information.
- Complete three-dimensional enumeration and certified non-homeomorphic pairs remain unfinished.

7 Finite Verification Plan

Start with one tetrahedron and a fixed pairing of its four faces. Each pair admits six affine gluing maps, giving

$$6^2 = 36$$

labeled assignments.

The next case uses a fixed two-tetrahedron graph with four paired pairs of faces, giving

$$6^4 = 1296$$

assignments before filtering and deduplication.

The verification pipeline is:

fixed face-pairing graph
$\downarrow$
enumeration of gluing maps
$\downarrow$
local topology and orientability checks
$\downarrow$
combinatorial deduplication
$\downarrow$
integral homology and stronger invariants
$\downarrow$
certified conclusions or unresolved pairs.

Different triangulation signatures alone do not prove non-homeomorphism. Equal selected invariants alone do not prove homeomorphism.

8 Seven-Week Plan

Week	Main Task
1	Finalize the scope and connect the literature review to the fixed-graph question.
2	Implement and validate gluing data and vertex-link calculations.
3	Complete the 36 one-tetrahedron assignments and their integral homology analysis.
4	Extend to a fixed two-tetrahedron graph and record boundary-state histories.
5	Analyze candidate pairs and separate proved results from unresolved cases.
6	Complete the final report draft and figures.
7	Reproduce the core results, revise the report, and prepare the presentation.

9 Expected Outcome

The expected outcome is a small, reproducible study showing how fixed combinatorial data can support different constructions and how selected invariants distinguish them.

The report will clearly separate known theorems, reconstructed arguments, finite computations, numerical observations, and working hypotheses. A new general classification theorem is not required for completion.

References

[1] X. Liu and S. Zhou, *Eigenvalues of Cayley graphs*, The Electronic Journal of Combinatorics 29(2) (2022), Paper P2.9. https://arxiv.org/abs/1809.09829.

[2] G. Brinkmann, *A Practical Algorithm for the Computation of the Genus.* https://arxiv.org/abs/2005.08243.

[3] S. Barthel and F. Buccoliero, *Constructing embedded surfaces for cellular embeddings of leveled spatial graphs.* https://arxiv.org/abs/2406.03800.

[4] B. A. Burton and W. Pettersson, *Fixed Parameter Tractable Algorithms in Combinatorial Topology*, arXiv:1402.3876v2. https://arxiv.org/abs/1402.3876v2.